

Development of a Deep Reinforcement Learning Agent for Optimal Project Portfolio Budgeting under Uncertain Cash Flows

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Abstract

Large contracted project portfolios expose governing boards to a recurring operational decision absent from the formal literature: how to allocate budget across active projects at each discrete period to maximise discounted net cash flow under cash-flow uncertainty, FIDIC contract mechanics, and the options to intervene in or terminate underperforming projects. This paper addresses that gap through a three-level modelling arc. **Level 1** formulates the problem as a deterministic full-foresight Mixed-Integer Linear Program whose optimum constitutes a provable upper bound on any causal policy operating under uncertainty. **Level 2** reintroduces stochastic contractor performance, establishing that the resulting multi-stage stochastic program is computationally intractable at realistic portfolio scales due to exponential scenario-tree growth. **Level 3** reformulates the problem as a Constrained Markov Decision Process, implemented as a configurable Gymnasium-compatible environment calibrated from anonymised EPC contractor data, on which a Proximal Policy Optimisation agent is trained and evaluated against the Level 1 oracle, a two-stage stochastic program, and a rule-based heuristic. The environment itself—encoding advance payment recovery, milestone-based payments, retention release, symmetric deviation penalties, and a two-condition termination gate—constitutes the primary contribution.

Keywords: project portfolio management; budget allocation; reinforcement learning; constrained Markov decision process; earned value management; FIDIC contracts; stochastic optimisation; mixed-integer linear programming.

1 Introduction

2 Introduction

2.1 Motivation and Problem Context

Large infrastructure and engineering projects routinely overrun their budgets and miss their schedules. Across a database of thousands of projects spanning seven decades, ? document that cost overruns are the norm rather than the exception, with mean overruns exceeding 20% in transportation infrastructure and reaching 70% in large-scale IT systems. ? reports that fewer than 40% of industrial megaprojects achieve their approved cost

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and schedule targets. These figures describe individual projects. When a portfolio of such projects is executed simultaneously under a single governing body and a shared capital commitment, the risks compound: a cash-flow shortfall on one project propagates immediately to others, and the decision of how much budget to direct to each project at each period can determine whether the portfolio as a whole succeeds or fails.

Despite this, the formal treatment of the portfolio-level budget allocation problem as a sequential decision-making problem is conspicuously absent from the project portfolio management literature. Research attention has been concentrated at the *selection* stage — which projects to include in the portfolio — while the *operational governance* stage — how to manage the portfolio of committed projects through to completion — has received comparatively little analytical treatment (???). The present work fills this gap.

We consider a portfolio of n projects, each governed by a bilateral contract consistent with FIDIC forms (?), executing concurrently under a fixed capital commitment. A portfolio manager — a board, investment committee, or programme director — exercises a single recurring decision at each discrete planning period: how much budget to allocate to each active project. All strategic and operational outcomes of interest flow through this allocation decision. The formal problem is to find the allocation policy that maximises discounted net cash flow across the portfolio horizon while satisfying hard liquidity constraints, respecting contract mechanics, and managing the options to intervene in underperforming projects or terminate them entirely.

2.2 The Project Management Decision Hierarchy

To situate the present contribution precisely, it is useful to distinguish the three nested decision layers that constitute project portfolio management (?).

Portfolio selection. The selection problem asks: given a candidate set of projects and a capital budget, which subset should the organisation commit to? This is a combinatorial optimisation problem studied extensively in the literature using mean-variance portfolio theory (??), multi-criteria decision analysis (?), and metaheuristic methods (?). The present work treats selection as an exogenous input: the portfolio of committed projects is fixed at the start of the planning horizon, and contracts have already been signed. This decomposition is consistent with the two-stage governance model of ? and the PMI Standard for Portfolio Management (?).

Portfolio governance. Once a portfolio is committed, the governing body must monitor and control execution. This is the layer at which the present work operates. The governing body does not manage individual projects directly — that is the contractor’s role — but it does control the budget directed to each project at each period, and it holds two options: triggering a management intervention to accelerate recovery on an underperforming project, and exercising the contractual termination clause if a project’s performance has deteriorated beyond recoverable bounds. These three instruments — allocation, intervention, and termination — constitute the full action space of the portfolio manager.

Project execution. At the project level, the contractor controls workforce, materials, equipment, and subcontracting. The portfolio manager observes the outcomes of these decisions — cumulative progress, earned value, payment certifications — but does not control them. The efficiency with which a budget allocation is converted into physical

progress is determined at the project level and is uncertain from the portfolio manager’s perspective.

This hierarchy is central to the modelling architecture: the formulation developed in Section ?? lives entirely at the portfolio governance layer, with project execution dynamics appearing only as parameters (deterministic at Level 1, stochastic at Levels 2 and 3).

2.3 Contract Mechanics and Cash-Flow Structure

The financial interaction between the portfolio manager (as client) and each contractor is governed by the contract. This paper parameterises all contract mechanics consistently with FIDIC Red, Yellow, and Silver Book forms (?), which represent the dominant international framework for EPC contracting. Understanding the cash-flow structure is essential to understanding the optimisation problem; this subsection provides that foundation.

Milestone-based payment. Rather than paying contractors on a time-basis, FIDIC contracts structure payment around certified milestones: discrete progress thresholds at which the client verifies that a specified scope fraction has been physically completed and releases the corresponding payment. The gross payment at milestone j is a contractually fixed fraction $\phi_{i,j}$ of the contract price CP_i . Because payment is contingent on certified progress, the timing of cash inflows is a function of the contractor’s actual performance and, through the portfolio manager’s allocation decisions, of the budget directed to the project. This creates the fundamental linkage between allocation decisions and cash inflows that makes the problem non-trivial.

Advance payment and progressive recovery. At project initiation, the client disburses an advance payment $A_i = \alpha_i \cdot CP_i$ to the contractor. This is a timing instrument: it reduces the contractor’s early working-capital burden and is recovered progressively through deductions from subsequent milestone payments. The recovery is proportional to milestone payment weights, ensuring the full advance is recovered by final completion. The advance payment creates an immediate cash outflow at project start and a series of reduced inflows at each milestone, both of which must be tracked in the portfolio cash balance.

Retention. A fraction ρ_i of each milestone payment is withheld by the client as a performance bond. The accumulated retention $R_i^{\text{ret}} = \rho_i \cdot CP_i$ is released in full upon project completion. Retention creates a structural completion incentive embedded in the payment mechanics: the larger the retention fraction, the more strongly the cash flow profile rewards completion over partial delivery. In the formulation, retention release enters the portfolio cash balance as a lump inflow in the period the final milestone is certified.

Earned value management as a performance signal. The Earned Value Management framework (?) provides the diagnostic vocabulary used to monitor contractor performance. The Cost Performance Index $CPI_{i,t} = BCWP_{i,t}/ACWP_{i,t}$ measures how much budgeted value has been earned per unit of actual expenditure; values below unity indicate cost overrun. The Schedule Performance Index $SPI_{i,t} = BCWP_{i,t}/BCWS_{i,t}$ measures actual progress relative to planned progress; values below unity indicate schedule lag (?). In the present formulation, these indices are observable state variables that inform the allocation decision; their stochastic dynamics at Levels 2 and 3 are calibrated from empirical EVM datasets (??).

S-curve expenditure profiles. The planned expenditure of a construction or engineering project follows a characteristic sigmoid profile over its planned duration: slow ramp-up in the mobilisation phase, rapid expenditure in the peak execution phase, and tapering in the close-out phase (?). This profile is modelled using the regularized incomplete Beta function $F_i(\tau) = I_\tau(a_i, b_i)$, whose shape parameters (a_i, b_i) are calibrated from project data. The planned cumulative expenditure fraction at normalised time $\tau_{i,t}$ provides the baseline against which actual progress is measured throughout the formulation.

Termination. FIDIC contracts provide for client-initiated termination when the contractor has materially breached the contract — most commonly through sustained failure to meet schedule obligations. Termination triggers a settlement payment based on the value of work completed at the termination date, net of any outstanding advance recovery obligation. In the present model, termination is a deliberate decision by the portfolio manager, not an exogenous event: it is triggered by the joint satisfaction of a cumulative penalty threshold and a sustained allocation floor, and its timing affects the portfolio cash balance through the settlement payment R_i^{term} , which may be positive or negative depending on the advance recovery position. Real options theory (??) provides the theoretical basis for treating termination as the exercise of an abandonment option.

2.4 The Three-Level Modelling Arc

The formulation of Section ?? proceeds across three levels of increasing generality. This subsection provides the conceptual roadmap; each level is developed formally in its corresponding subsection.

Level 1: Deterministic full-foresight MILP. The natural starting point for any optimisation problem is its deterministic version: assume all uncertain parameters are known, and find the optimal decision sequence. In the portfolio budget allocation problem, the key uncertain parameter is the budget-to-progress efficiency $\eta_{i,t}$ — the scalar that determines how much physical progress a given budget allocation yields in each period. If $\eta_{i,t}$ is treated as a known constant $\bar{\eta}_{i,t}$, the progress recursion becomes a deterministic linear map from budget allocations to physical outcomes, and the entire optimisation problem collapses to a Mixed-Integer Linear Program (MILP).

The MILP is not proposed as a practical solution method — no real portfolio manager has advance knowledge of contractor efficiency trajectories. Rather, it serves two purposes. First, its optimal value Z_{L1}^* is a provable upper bound on the performance of any causal policy operating under uncertainty: since the MILP optimises with perfect foresight, it can do no worse than any policy that must make decisions without knowing future outcomes. Second, it provides an LP foundation from which the stochastic extensions of Levels 2 and 3 are constructed by systematic relaxation of the deterministic assumption.

The MILP requires two structural choices that are standard in combinatorial optimisation. Milestone certification is a binary event — a milestone is either certified or not — captured by introducing binary decision variables $u_{i,j,t} \in \{0, 1\}$. The symmetric running deviation penalty involves an absolute value and a cap, both of which are linearised by introducing auxiliary variables $d_{i,t}^+, d_{i,t}^- \geq 0$ for the positive and negative parts of the deviation. These linearisations are exact: no approximation is introduced.

Level 2: Stochastic extension and intractability. When $\eta_{i,t}$ is replaced by a random variable — as it must be to reflect reality — the MILP becomes a multi-stage stochastic

program (?). The scenario-tree representation of this program grows exponentially: if $\eta_{i,t}$ takes K discrete values at each of H periods for each of n projects, the scenario tree has K^{nH} leaf nodes. For a portfolio of ten projects over a 24-month horizon with five efficiency scenarios per project per period, this is 5^{240} leaf nodes — a number that renders exact stochastic programming computationally intractable. This is not merely a computational inconvenience; it is the fundamental barrier that motivates the Level 3 reformulation.

Level 3: Constrained MDP and RL agent. The Constrained Markov Decision Process (?) provides the tractable reformulation. Rather than optimising over all scenario-tree paths simultaneously, the CMDP formalises the problem as a sequential interaction between a policy and an environment: at each period, the policy observes the current portfolio state, selects a budget allocation, and the environment transitions stochastically to the next state. The key insight is that the CMDP does not require explicit scenario enumeration: the policy is optimised through repeated interaction with the environment, using sampled trajectories rather than exhaustive scenario trees.

The mapping from MILP to CMDP is exact in structure. The decision variables $\{x_{i,t}\}$ become the action space; the state variables $(P_{i,t}^{\text{actual}}, E_{i,t}, \text{CPI}_{i,t}, \text{SPI}_{i,t}, \delta_{i,t}, \Delta_{i,t}^{\text{pen}}, \text{status}_{i,t}, B_t)$ become the observation space; the budget constraint becomes the constrained action space; and the objective function becomes the reward signal. The CMDP is implemented as a configurable Gymnasium-compatible environment (?), and a Proximal Policy Optimisation agent (?) is trained on it using domain randomisation (?) to ensure generalisation across project types and contract structures.

2.5 The Operations Research Lineage

The present work is situated squarely within the Industrial Engineering and Operations Research tradition. The mathematical tools employed — mixed-integer linear programming, stochastic programming, dynamic programming, and Markov decision processes — share a common intellectual lineage that traces directly to Dantzig’s simplex method (?), Bellman’s principle of optimality (?), and the stochastic programming framework of ?.

The MDP formalisation used in Level 3 is, mathematically, a computationally tractable approach to solving a Bellman equation in a high-dimensional state space. The state space in the present problem — $7n + 1$ continuous and categorical variables for a portfolio of n projects — is too large for tabular dynamic programming and too structured for model-free value-iteration methods, but well-suited to policy-gradient methods that learn a parameterised policy directly. PPO (?) is the policy-gradient algorithm of choice for its stability properties in continuous action spaces and its established use in operations management domains (??).

The reinforcement learning agent is therefore not presented as a departure from the OR tradition, but as its continuation into the regime where the curse of dimensionality makes analytical dynamic programming intractable. The Level 1 MILP and Level 2 stochastic program are not straw men; they are the canonical OR solutions to the deterministic and stochastic versions of the problem, respectively, and both serve as baselines against which the Level 3 agent is evaluated.

2.6 Practical Deployment Context

Incumbent project management information systems — most prominently Oracle Primavera P6 and Microsoft Project — provide rich project-level schedule and cost tracking but lack a portfolio-level optimisation layer. The portfolio manager’s recurring budget

allocation decision is currently supported, at best, by spreadsheet models, earned value dashboards, and expert judgement. The environment developed in Level 3 constitutes the analytical engine that such a portfolio-level decision-support module would require: it integrates contract mechanics, performance uncertainty, and multi-project cash-flow dynamics in a single configurable framework. This positions the contribution not only as a theoretical advance but as a step toward a deployable module for portfolio management information systems, a direction explicitly identified as a priority in the project management software literature (?).

2.7 Paper Organisation

The remainder of the paper is organised as follows. Section ?? reviews the four literature streams on which the work draws: project management foundations, project portfolio management and selection, PPM budgeting and cash-flow modelling, and reinforcement learning for sequential resource allocation. Section ?? develops the three-level problem formulation: the deterministic MILP (Level 1), the stochastic extension and intractability argument (Level 2), and the CMDP reformulation and Gymnasium-compatible environment (Level 3). Section ?? presents the empirical evaluation: calibration of environment parameters, training protocol, and comparison of the PPO agent against the Level 1 MILP oracle, a two-stage stochastic program, and a rule-based heuristic. Section ?? concludes with implications for portfolio management practice, limitations of the framework, and directions for future research including integration with incumbent PMIS tools.

3 Problem Formulation

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4 Problem Formulation

This section develops the formal problem structure for portfolio-level budget allocation in contracted project portfolios. The formulation proceeds across three levels. **Level 1** (Section ??) treats all contractor performance parameters as known and establishes the deterministic MILP whose optimum constitutes a theoretical upper bound: no implementable policy operating under uncertainty can exceed it. **Level 2** (Section ??) reintroduces stochastic efficiency and characterizes the resulting multi-stage stochastic program, establishing that its scenario tree grows exponentially in the number of projects and periods. **Level 3** (Section ??) reformulates the intractable stochastic program as a Constrained Markov Decision Process (CMDP), providing an explicit mapping from LP elements to MDP primitives and yielding a Gymnasium-compatible environment on which the PPO agent of Section ?? is trained. The three levels are a coherent arc — Level 1 defines the ceiling, Level 2 explains why it is unreachable, Level 3 provides the tractable path — all sharing the primitives defined next.

4.1 Scope, Assumptions, and Problem Primitives

Scope. The problem is operational, not strategic: the portfolio of contracted projects is fixed, commitments are signed, and the decision-maker exercises a single lever — the budget $x_{i,t}$ directed to each active project i at each discrete period t . All outcomes of interest (milestone achievement, cash balance, schedule adherence, termination) flow through this allocation decision. Portfolio selection is treated as exogenous ?.

Structural assumptions. Four choices underlie the entire formulation. (i) *Independent bilateral contracts*: projects interact only through the shared budget constraint; no resource or workforce sharing occurs, consistent with standard EPC contracting practice. (ii) *Budget as the sole control*: the contractor’s internal resource decisions are absorbed into the efficiency parameter $\eta_{i,t}$; the portfolio manager observes only how much physical progress a budget allocation yields, not how it is achieved. (iii) *Discrete monthly periods*: $\mathcal{T} = \{1, \dots, H\}$ reflects the monthly governance cycle of portfolio management (progress reporting, payment certification, performance review). (iv) *FIDIC contract mechanics*: the milestone payment structure, advance payment and recovery, retention, and termination settlement are parameterized consistently with FIDIC Red, Yellow, and Silver Book forms ?; all contract-specific quantities are calibrated parameters, not fixed constants.

4.1.1 Sets, Indices, and the Decision Variable

[leftmargin=2.2cm, labelwidth=2.0cm]Projects $\{1, \dots, n\}$, indexed by i ; all committed at horizon start. Planning periods $\{1, \dots, H\}$, indexed by t (calendar months). Ordered milestones $\{1, \dots, M_i\}$ for project i , indexed by j , with thresholds $\theta_{i,1} < \dots < \theta_{i,M_i} = 1$. Planned execution window $\{s_i, \dots, f_i\}$; anchors S-curve computations. Projects active at period t ; fixed parameter at Level 1, endogenous at Levels 2–3. Milestones of project i certified by period t : $\{j : \theta_{i,j} \leq P_{i,t}^{\text{actual}}\}$. **Decision variable**: budget allocated to project i at period t . The period- t action is $\mathbf{x}_t = (x_{i,t})_{i \in \mathcal{P}_t^{\text{active}}}$. Portfolio cash balance at the start of period t ; B_1 is given.

4.1.2 Project Parameters

Tables ?? and ?? list all project parameters, fixed at commitment. The planned cumulative expenditure fraction follows a normalized S-curve ?? parameterized by (a_i, b_i) :

$$P_{i,t}^{\text{plan}} = F_i(\tau_{i,t}^c) = I_{\tau_{i,t}^c}(a_i, b_i), \quad \tau_{i,t} = \frac{t - s_i + 1}{D_i^{\text{plan}}}, \quad \tau_{i,t}^c = \min\{1, \max\{0, \tau_{i,t}\}\},$$

where $I_\tau(\cdot, \cdot)$ is the regularized incomplete Beta function. The planned period expenditure increment is $\Delta \bar{E}_{i,t} = \text{BAC}_i [F_i(\tau_{i,t}^c) - F_i(\tau_{i,t-1}^c)]$. This profile is a fixed parameter derived at commitment from $(a_i, b_i, s_i, f_i, \text{BAC}_i)$; it is never a decision variable.

Parameter	Symbol	Domain	Definition
Planned start & finish Schedule	s_i, f_i	$\mathbb{Z}^+$	Anchor of execution window $\mathcal{A}_i^{\text{plan}}$ and S-curve.
Planned duration Schedule	D_i^{plan}	$\mathbb{Z}^+$	$f_i - s_i + 1$; normalizes progress against plan.
S-curve shape Schedule	a_i, b_i	$\mathbb{R}^+$	Beta CDF parameters governing the planned expenditure profile.
Milestone threshold Schedule	$\theta_{i,j}$	$(0, 1]$	Cumulative progress fraction required to certify milestone j .
Delay buffer Schedule	buffer_i	$\mathbb{Z}_0^+$	Periods beyond f_i before forced termination is assessed.
Contract price Contract	CP_i	$\mathbb{R}^+$	Total client payment; $CP_i = \text{BAC}_i(1 + \pi_i)$.
Profit margin Contract	π_i	$\mathbb{R}_0^+$	Contractual markup on BAC_i .
Advance payment ratio Contract	α_i	$[0, 1]$	$A_i = \alpha_i \cdot CP_i$ disbursed at $t = s_i$; recovered via milestone deductions.
Retention ratio Contract	ρ_i	$[0, 1]$	Fraction withheld per milestone; released as $R_i^{\text{ret}} = \rho_i \cdot CP_i$ at completion.
Milestone weight Contract	$\phi_{i,j}$	$\mathbb{R}^+$	Fraction of CP_i at milestone j ; $\sum_j \phi_{i,j} = 1$.

Table 1: Schedule and contract parameters for project $i \in \mathcal{P}$ (fixed at commitment).

Parameter	Symbol	Domain	Definition
Budget at completion Cost	BAC_i	$\mathbb{R}^+$	Total planned contractor expenditure; EVM cost basis.
Discount factor Cost	γ	$(0, 1]$	Per-period discount applied to future cash flows; shared across $\mathcal{P}$.
Initial cash balance Cost	B_1	$\mathbb{R}^+$	Portfolio cash balance at horizon start; initial condition of balance recursion.
Deviation penalty weight Penalty	λ_i	$\mathbb{R}_0^+$	Per-period cost per unit absolute progress deviation; calibrated so cumulative penalty at planned finish under sustained contractual-limit delay equals $\Lambda_i \cdot CP_i$.
Deviation penalty cap Penalty	Λ_i	$\mathbb{R}_0^+$	Maximum cumulative penalty as fraction of CP_i ; applies symmetrically to lag and acceleration.
Termination threshold Termination	κ_i	$(0, 1]$	Fraction of $\Lambda_i \cdot CP_i$ at which project enters termination candidacy.
Minimum allocation Termination	m_i^0	$\mathbb{R}_0^+$	Floor budget to hold project in contractually serviceable state while termination conditions are evaluated; no performance-recovery role.
Confirmation window Termination	k	$\mathbb{Z}^+$	Consecutive periods at floor m_i^0 required to confirm abandonment.

Table 2: Cost, penalty, and termination parameters for project $i \in \mathcal{P}$ (fixed at commitment).

4.1.3 State Variables and Transition Dynamics

Project-level state. For each $i \in \mathcal{P}$, $t \in \mathcal{T}$: $P_{i,t}^{\text{actual}} \in [0, 1]$ (cumulative progress), $E_{i,t} \geq 0$ (cumulative expenditure), $SPI_{i,t} = P_{i,t}^{\text{actual}}/P_{i,t}^{\text{plan}}$, $CPI_{i,t} = P_{i,t}^{\text{actual}} \cdot BAC_i/E_{i,t}$, $\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}}$ (signed deviation), $\Delta_{i,t}^{\text{pen}} \geq 0$ (cumulative penalty), and $\text{status}_{i,t} \in \{\text{active}, \text{completed}, \text{terminated}\}$. The portfolio state at period t is:

$$S_t = \left(\{P_{i,t}^{\text{actual}}, E_{i,t}, SPI_{i,t}, CPI_{i,t}, \delta_{i,t}, \Delta_{i,t}^{\text{pen}}, \text{status}_{i,t}\}_{i \in \mathcal{P}}, B_t \right).$$

Budget-to-progress conversion. Budget $x_{i,t}$ converts to progress through the efficiency $\eta_{i,t}$:

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \eta_{i,t} \cdot \frac{x_{i,t}}{BAC_i}, \quad E_{i,t} = E_{i,t-1} + x_{i,t}, \quad (1)$$

with $P_{i,0}^{\text{actual}} = E_{i,0} = 0$. The scalar $\eta_{i,t}$ absorbs all sources of conversion uncertainty (productivity, supply chain, weather). At Level 1, $\eta_{i,t}$ is replaced by the known deterministic value $\bar{\eta}_{i,t}$ (equal to unity under nominal plan); at Levels 2–3 it is a stochastic primitive realized only after $x_{i,t}$ is committed.

Contractual cash-flow mechanics. Payment inflows follow FIDIC mechanics ?. At initiation ($t = s_i$), the advance payment $A_i = \alpha_i \cdot CP_i$ is received. Milestone j is certified

at the first t with $P_{i,t}^{\text{actual}} \geq \theta_{i,j}$; the net payment (after retention and advance recovery) is:

$$R_{i,j}^{\text{net}} = \phi_{i,j} \cdot CP_i(1 - \rho_i) - \phi_{i,j} \cdot A_i. \quad (2)$$

Upon completion ($P_{i,t}^{\text{actual}} = 1$) the accumulated retention $R_i^{\text{ret}} = \rho_i \cdot CP_i$ is released together with the final milestone payment. The payment identity $A_i + \sum_j R_{i,j}^{\text{net}} + R_i^{\text{ret}} = CP_i$ holds exactly. Aggregate period inflow is:

$$V_t = \sum_{i,j} R_{i,j}^{\text{net}} \cdot \mathbf{1}\{(i,j) \text{ newly certified at } t\} + \sum_{i \in \mathcal{P}_t^{\text{term}}} R_i^{\text{term}}, \quad (3)$$

where $\mathcal{P}_t^{\text{term}}$ collects projects newly terminated at t , with settlement R_i^{term} defined in (??) below.

Symmetric running deviation penalty. Schedule adherence is priced at every period:

$$c_{i,t}^{\text{dev}} = \min\{\lambda_i |\delta_{i,t}|, \Lambda_i \cdot CP_i\}, \quad \Delta_{i,t}^{\text{pen}} = \sum_{\tau=s_i}^t c_{i,\tau}^{\text{dev}}. \quad (4)$$

The penalty is symmetric ($|\delta_{i,t}|$, not $\max\{0, -\delta_{i,t}\}$): both lag and acceleration are penalized, discouraging budget-pushing that risks quality degradation ?. The cap $\Lambda_i \cdot CP_i$ mirrors the contractual delay-penalty ceiling under FIDIC terms.

Termination conditions and settlement. A project transitions to **terminated** at the first period t at which both conditions hold simultaneously:

$$(i) \text{ Penalty saturation: } \Delta_{i,t}^{\text{pen}} \geq \kappa_i \cdot \Lambda_i \cdot CP_i, \quad (5)$$

$$(ii) \text{ Confirmed floor: } x_{i,\tilde{t}} \leq m_i^0 \quad \forall \tilde{t} \in \{t - k + 1, \dots, t\}. \quad (6)$$

Condition (??) signals exhaustion of contractual schedule tolerance; condition (??) confirms that the portfolio manager has deliberately ceased investment rather than transiently reduced it. Upon termination at t_i^{term} , the net settlement is:

$$R_i^{\text{term}} = P_{i,t_i^{\text{term}}}^{\text{actual}} \cdot CP_i(1 - \rho_i) - \left(A_i - \sum_{j \in \mathcal{M}_i^{\text{cert}}(t_i^{\text{term}})} \phi_{i,j} \cdot A_i \right). \quad (7)$$

Negative R_i^{term} (outstanding advance exceeds earned value) constitutes a net obligation recovered from portfolio inflows, consistent with FIDIC termination clauses.

Portfolio cash balance recursion.

$$B_{t+1} = B_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} + V_t, \quad B_t \geq 0 \quad \forall t \in \mathcal{T}. \quad (8)$$

The constraint $B_t \geq 0$ prohibits borrowing; the portfolio operates under a fixed capital commitment ?. The shared balance B_t is the sole coupling across otherwise independent project dynamics, and its linear scaling with n characterizes the complexity that motivates the intractability argument in Level 2.

4.1.4 Shared Objective

The decision-maker maximizes discounted net cash flow over the horizon:

$$\max_{\{x_{i,t}\}} \mathbb{E} \left[\sum_{t=1}^H \gamma^{t-1} \left(V_t - \sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} - \sum_{i \in \mathcal{P}_t^{\text{active}}} c_{i,t}^{\text{dev}} \right) \right], \quad (9)$$

subject to, for all $i \in \mathcal{P}$, $t \in \mathcal{T}$:

$$x_{i,t} \geq 0, \quad (10)$$

$$\sum_{i \in \mathcal{P}_t^{\text{active}}} x_{i,t} \leq B_t, \quad (11)$$

$$x_{i,t} \geq m_i^0 \cdot \mathbf{1}\{\text{status}_{i,t} = \text{active}\}. \quad (12)$$

There is no terminal sum. Completion incentives are embedded structurally: R_i^{ret} and R_{i,M_i}^{net} enter V_t at the certification period; R_i^{term} (possibly negative) enters at termination, creating a natural deterrent against premature abandonment. Constraint (??) ensures each active project remains in a contractually serviceable state; it is vacuous once $\text{status}_{i,t} \neq \text{active}$. The expectation $\mathbb{E}[\cdot]$ collapses to a deterministic evaluation at Level 1 and is retained at Levels 2 and 3.

4.2 Level 1: Deterministic Full-Foresight MILP

Setting $\eta_{i,t} = \bar{\eta}_{i,t}$ (known) renders S_t a deterministic function of $\mathbf{x}_t$, so $\mathcal{P}_t^{\text{active}}$, milestone certification times, and termination periods t_i^{term} are all computable ex ante. Introducing the certification binary $u_{i,j,t} \in \{0,1\}$ ($= 1$ iff milestone j of project i certified by period t) and activity indicator $\text{act}_{i,t} \in \{0,1\}$ (fixed parameter), the stochastic objective (??) collapses to the MILP:

$$\max \sum_{t=1}^H \gamma^{t-1} \left(V_t^{\text{det}} - \sum_{i \in \mathcal{P}} \text{act}_{i,t} x_{i,t} - \sum_{i \in \mathcal{P}} \text{act}_{i,t} c_{i,t}^{\text{dev}} \right) \quad (13a)$$

$$\text{s.t. } B_{t+1} = B_t - \sum_i \text{act}_{i,t} x_{i,t} + V_t^{\text{det}}, \quad B_t \geq 0, \quad \forall t \quad (13b)$$

$$\sum_i \text{act}_{i,t} x_{i,t} \leq B_t, \quad \forall t \quad (13c)$$

$$P_{i,t}^{\text{actual}} = P_{i,t-1}^{\text{actual}} + \bar{\eta}_{i,t} \frac{x_{i,t}}{\text{BAC}_i}, \quad P_{i,t}^{\text{actual}} \leq 1, \quad \forall i, t \quad (13d)$$

$$P_{i,t}^{\text{actual}} \geq \theta_{i,j} u_{i,j,t} - \mathbf{M}(1 - u_{i,j,t}), \quad \forall i, j, t \quad (13e)$$

$$P_{i,t}^{\text{actual}} \leq \theta_{i,j} + \mathbf{M} u_{i,j,t}, \quad u_{i,j,t} \geq u_{i,j,t-1}, \quad u_{i,j+1,t} \leq u_{i,j,t}, \quad \forall i, j, t \quad (13f)$$

$$\begin{aligned} V_t^{\text{det}} = & \sum_{i,j} R_{i,j}^{\text{net}} (u_{i,j,t} - u_{i,j,t-1}) + \sum_i R_i^{\text{ret}} (u_{i,M_i,t} - u_{i,M_i,t-1}) \\ & + \sum_i A_i \cdot \mathbf{1}\{t=s_i\} + \sum_{i \in \mathcal{P}^{\text{term}}} R_i^{\text{term}} \cdot \mathbf{1}\{t=t_i^{\text{term}}\}, \quad \forall t \end{aligned} \quad (13g)$$

$$\delta_{i,t} = P_{i,t}^{\text{actual}} - P_{i,t}^{\text{plan}} = d_{i,t}^+ - d_{i,t}^-, \quad d_{i,t}^+, d_{i,t}^- \geq 0, \quad \forall i, t \quad (13h)$$

$$c_{i,t}^{\text{dev}} \geq \lambda_i (d_{i,t}^+ + d_{i,t}^-), \quad c_{i,t}^{\text{dev}} \leq \Lambda_i \cdot CP_i, \quad c_{i,t}^{\text{dev}} \geq 0, \quad \forall i, t \quad (13i)$$

$$x_{i,t} \geq m_i^0 \cdot \text{act}_{i,t}, \quad \forall i, t \quad (13j)$$

$$x_{i,t} \geq 0, \quad u_{i,j,t} \in \{0,1\}, \quad u_{i,j,0} = P_{i,0}^{\text{actual}} = 0. \quad (13k)$$

The big- M constant in (??)–(??) is $\mathbf{M} = \max_{i,j} \theta_{i,j}$; tighter values may be derived project-specifically. Constraints (??)–(??) implement the exact if-and-only-if certification logic with monotone, ordered binary activation. Constraints (??)–(??) linearize the absolute-value and cap in (??): the maximizing objective drives $c_{i,t}^{\text{dev}}$ to the tighter of the two bounds, recovering the exact symmetric penalty. The difference $(u_{i,j,t} - u_{i,j,t-1})$ in (??) equals one in the sole period a milestone is first certified, linearizing the trigger without additional binaries. Termination timing t_i^{term} and the settlement R_i^{term} (??) are pre-computed from $\bar{\eta}_{i,t}$ before solving; the termination conditions (??)–(??) are parameter-verifiable at Level 1 and become binding optimization constraints at Levels 2–3.

Upper bound. Let Z_{L1}^* denote the Level 1 optimum. For any causal policy π feasible under uncertainty, $\sup_{\pi} \mathbb{E}[Z^{\pi}] \leq Z_{\text{L1}}^*$: every feasible causal allocation sequence is also feasible in the Level 1 program solved on the realized trajectory, so the full-foresight optimum dominates in expectation. The gap $Z_{\text{L1}}^* - Z^{\pi}$ quantifies the cost of uncertainty and anchors the empirical comparison in Section ??.